作业4

2.5.5 (1) $V_{CEQ}=\frac{V_{CC}}{2}=15V$ $I_{CQ}=\frac{V_{CE}-\frac{V_{CE}}{2}}{R_C}=10mA$

(2) $V_{om}=\frac{V_{CC}}{2}-|V_{CES}|=12V$

$P_{om}=\frac{V_{om}^2}{2R_L}=9W$

$P_E=\frac{V_{CC}}{2}\cdot\int_0^{\pi}\frac{i_L}{\pi}d\omega t=\frac{V_{CC}V_{om}}{\pi R_L}=14.3W$

$\therefore \eta=\frac{9}{14.3}\times 100\%=62.8\%$

2.5.6 (1) $A_{vf}=-\frac{R_2}{R_1}=-10$

(2) A最大输出电压幅度±10V $\therefore V_{om}=\pm 9.3V$

$i_L=\frac{V_o}{R_L}=1.1625A$

$\therefore i_C=i_L+\frac{V_o}{R_2}=1.1625A$ $\beta=\frac{i_C}{10mA}=116.25$

$\therefore \beta$至少117

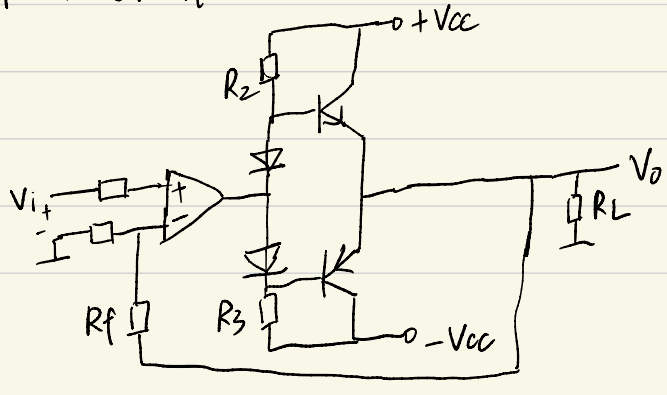
(3) $P_{om}=\frac{V_{om}^2}{2R_L}=5.41W$ (4) 电源供给功率 $P_E=V_{CC}\int_0^{\pi}\frac{i_C}{\pi}d\omega t$

$=11.1W$

$\eta=\frac{P_{om}}{P_E}\times 100\%=48.7\%$

$P_{T2}=2.85W$

2.5.7 (1) 电压串联负反馈



(2) $V_o = 5V$ $V_i = \frac{R_1}{R_1+R_f} V_o$ $\therefore R_f = 49k\Omega$

(3) $V_{AV} = \pm 13V$ $V_o = V_{AV} = \pm 13V$

$P_{om} = \frac{V_{om}^2}{2R_L} = 10.56W$

2.5.11 (1) $V_{O(AV)} = 0.9V_2 = 18V$

(2) $V_{O(AV)} = \sqrt{2}V_2 = 20\sqrt{2}V \approx 28.3V$

(3) 一个二极管开路，半周期整流桥全波

C充电至$\sqrt{2}V$

$\therefore V_{O(AV)} = \sqrt{2}V_2 \approx 28.3V$

(4) $V_{O(AV)} = 1.2V_2 = 24V$

2.5.12

(1) $R = \frac{1.2V_2 - V_o}{I_Z + I_L}$ $I_Z \geq 5mA$

$\therefore R \leq 413\Omega$ 可选$400\Omega$

(2) $I_{Zmax} = \frac{1.2V_2 - V_o}{R} - I_{Lmax} = 36.9mA$

2.5.14 (1) 上 (+)

下 (−)

(2) $I_{O(max)} \approx \beta I_{B(max)} = 100mA$

$V_{O(max)} = V_{AOmax} - V_{BE} = 19.3V$

$I_C = 100mA$ $V_{Omax} = 6V$

$\therefore P_{CM} = I_C V_{CEmax} = I_C(V_1 - V_{Omax}) = 1.8W$